MATHEMATICAL GAME THEORY

Midterm Examination

Fall 2024 · Prof. L. Nakamura

80 minutes · 60 points

Name: _____ ID: _____

Instructions. No notes, books, or electronic devices. Proofs should be complete but concise: state what you are assuming, cite results from lecture by name (e.g. "by the minimax theorem"), and make every inequality explicit. You may use the result of an earlier part in a later part even if you did not prove it.

Notation. $\Delta(S)$ is the set of probability distributions on a finite set S. For a mixed profile σ, $u_i(\sigma)$ is player i's expected payoff. In bimatrix tables the first entry belongs to the row player.

Problem	1	2	3	4	5	Total
Points	12	14	12	12	10	60
Score						

Problem 1 *Dominated strategies and best responses* (12 points)

Let $G = (N, (S_i)_{i \in N}, (u_i)_{i \in N})$ be a finite game. A pure strategy $s_i \in S_i$ is *strictly dominated* by $\tau_i \in \Delta(S_i)$ if $u_i(\tau_i, s_{-i}) > u_i(s_i, s_{-i})$ for every $s_{-i} \in S_{-i}$.

(a) [6] Prove that if s_i is strictly dominated by some $\tau_i \in \Delta(S_i)$, then $\sigma_i^*(s_i) = 0$ in every mixed-strategy Nash equilibrium σ^* of G.

(b) [3] Show by an explicit 2×2 example that the statement in (a) is false if "strictly dominated" is replaced by "weakly dominated".

(c) [3] In the game below only the row player's payoffs are shown. Let q be the probability that Column plays L. Show that M is not a best response to either pure strategy of Column, find the set of all q for which M is a best response, and decide whether M is strictly dominated.

	L	R
U	3	0
M	2	2
D	0	3

Problem 2 *Zero-sum games* (14 points)

Let A be an $m \times n$ real matrix, viewed as a zero-sum game in which Row chooses $x \in \Delta_m$, Column chooses $y \in \Delta_n$, and Column pays Row $x^\top A y$.

(a) [4] Prove directly (without the minimax theorem) that

$$\max_{x \in \Delta_m} \min_{y \in \Delta_n} x^\top A y \;\le\; \min_{y \in \Delta_n} \max_{x \in \Delta_m} x^\top A y.$$

(b) [6] Let $A = \operatorname{diag}(a_1, \dots, a_n)$ with every $a_i > 0$ (Row wins a_i if both players choose i, and nothing otherwise). Prove that the value of the game is

$$v = \Big(\sum_{i=1}^{n} \frac{1}{a_i} \Big)^{-1},$$

and that $x_i = y_i = v / a_i$ are optimal strategies for the two players.

(c) [4] Prove that Column's optimal strategy in (b) is unique. Then give v and the optimal strategies when $(a_1, a_2, a_3) = (2, 3, 6)$.

Problem 3 *Symmetric* 2×2 *games* (12 points)

A two-player game with strategy set $\{A, B\}$ for both players is *symmetric* if there is a function $u : \{A, B\}^2 \to \mathbb{R}$ such that the player choosing s against an opponent choosing t receives $u(s, t)$. Extend u to mixed strategies, writing $p \in [0, 1]$ for the probability of A.

(a) [7] Prove, without using Nash's existence theorem, that every symmetric 2×2 game has a *symmetric* Nash equilibrium (p^*, p^*).

(b) [5] Find all Nash equilibria (symmetric or not) of the symmetric game with $u(A, A) = 1$, $u(A, B) = 5$, $u(B, A) = 3$, $u(B, B) = 2$.

Problem 4 *Potential games* (12 points)

A finite game has an *exact potential* $\Phi : S \to \mathbb{R}$ if for every player i, every s_{-i}, and all $s_i, s_i' \in S_i$,

$$u_i(s_i', s_{-i}) - u_i(s_i, s_{-i}) = \Phi(s_i', s_{-i}) - \Phi(s_i, s_{-i}).$$

(a) [3] Prove that every finite game with an exact potential has a pure-strategy Nash equilibrium.

(b) [2] Prove that in such a game every sequence of strict unilateral improvements (each step, one player switches to a strategy that strictly raises her payoff) is finite.

(c) [7] Three commuters each choose road A or road B. If k commuters use a road, each of them incurs a travel cost $c_A(k) = 2k$ on road A or $c_B(k) = k + 3$ on road B; each commuter

minimizes her own cost. Let n_A and $n_B = 3 - n_A$ be the numbers of commuters on each road, and define

$$\Psi = \sum_{k=1}^{n_A} c_A(k) + \sum_{k=1}^{n_B} c_B(k).$$

Show that when a single commuter switches roads, the change in her cost equals the change in Ψ. Compute Ψ for $n_A = 0, 1, 2, 3$, find all pure Nash equilibria, and compare the equilibrium total cost with the minimum possible total cost.

Problem 5 *A divisor game* (10 points)

Fix an integer $N > 1$ and let S_0 be the set of positive divisors of N. Two players alternate moves, Player 1 first. The current position is a set $S \subseteq S_0$ (initially $S = S_0$). A move consists of choosing some $d \in S$ and replacing S by $S \setminus \{e \in S : d \mid e\}$ (i.e. removing d and all its multiples that remain). The player who chooses 1 loses. (Note that 1 remains available until someone chooses it.)

(a) [2] Explain why the game ends after finitely many moves and why exactly one of the players has a winning strategy.

(b) [5] Prove that Player 1 has a winning strategy for every $N > 1$. (*Hint:* consider Player 1 opening with N.)

(c) [3] Give an explicit winning strategy for Player 1 when $N = p^k$ for a prime p and $k \geq 1$, and when $N = pq$ for distinct primes p, q.

END OF EXAMINATION